

METAL VISION AI

Frequently Asked Questions — Industrial Metal Surface Inspection

1. What is Metal Vision AI?

It is an industrial surface inspection system that uses deep learning to classify metal surface defects from images.

2. Which model is used?

The system uses a ResNet50 transfer-learning model implemented with TensorFlow and Keras.

3. How many defect classes are supported?

Ten classes are supported: Punching Hole, Welding Line, Crescent Gap, Water Spot, Oil Spot, Silk Spot, Inclusion, Rolled Pit, Crease, and Waist Folding.

4. What happens during an inspection?

The image is uploaded, preprocessed, classified by ResNet50, and a Grad-CAM visualization is generated. The result is then stored in the inspection history.

5. What is Grad-CAM?

Grad-CAM highlights image regions that contributed strongly to the model's prediction.

6. Does Grad-CAM provide exact defect boundaries?

No. In this project it provides approximate defect-region localization. Exact bounding boxes require suitable annotations and a detection or segmentation model.

7. What input size does the model use?

Images are resized to 224 × 224 pixels before model inference.

8. How large is the dataset?

The project dataset contains 2,541 images across 10 defect classes, with a 70% training and 30% testing split.

9. Why are class weights used?

The dataset is imbalanced, so class weights help reduce bias toward majority classes during training.

10. Can the current model identify a GOOD or NORMAL surface?

No. The current classifier was trained only on the 10 defect classes. A genuine GOOD/NORMAL detector would require normal-surface training data and an appropriate binary or multi-class setup.

11. Where are previous inspections viewed?

The Inspection History page retrieves stored inspection records from the backend database.

12. Can a PDF report be generated?

Yes. A completed inspection can be exported as a PDF report from the result page.

13. What backend technology is used?

The backend uses Python and FastAPI. The frontend uses HTML, CSS, and JavaScript.

14. Is the trained model connected to the application?

Yes. FastAPI loads the trained ResNet50 model and provides the image-analysis API endpoint.

15. Which image formats are supported?

JPG, JPEG, and PNG images are supported.